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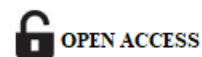


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FROM ENVIRONMENTAL CHALLENGES TO SUSTAINABLE SOLUTIONS: A COMPARATIVE ASSESSMENT OF RIVER POLLUTION CONTROL AND RESTORATION IN DEVELOPED AND DEVELOPING COUNTRIES

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ABSTRACT

River pollution remains a persistent environmental-management challenge associated with municipal wastewater, industrial effluents, agricultural runoff, solid waste, and other contaminant sources. This study comparatively examines river pollution control and restoration in four contrasting contexts: the Rhine River in Germany, the Thames River in the United Kingdom, the Ganga River in India, and the Citarum River in Indonesia. A comparative case-study approach was applied using secondary evidence from peer-reviewed literature, government agencies, river-basin organisations, and programme reports. Each case was assessed using a common framework covering pollution pressures, management interventions, institutional mechanisms, documented environmental outcomes, remaining pollution, and implementation limitations. The findings show measurable progress across all four cases, but with substantial differences in the form and extent of improvement. The Rhine experienced reductions in conventional pollutant loads alongside extensive ecological restoration, while contemporary management increasingly addresses micropollutants.

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The Thames demonstrates long-term water-quality recovery supported by wastewater infrastructure, although sewage-related pollution incidents continue. The Ganga shows improvement in selected water-quality indicators alongside major expansion of sewage-treatment capacity, while treatment and industrial-compliance gaps remain. The Citarum demonstrates continuing watershed-scale intervention, including wastewater control, waste removal, and industrial enforcement, but water-quality conditions remain spatially variable. Overall, the comparison indicates that river restoration depends on sustained pollution-source control, wastewater treatment, monitoring, regulatory enforcement, institutional coordination, and ecological restoration. The findings also demonstrate that management outputs do not necessarily correspond to uniform ecological recovery, highlighting the need for long-term, adaptive, and context-specific river-management strategies.

Keywords: Environmental Management, River Pollution, River Restoration, Sustainability, Water Quality

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1. INTRODUCTION

Rivers are essential components of freshwater systems, supporting domestic water supply, agriculture, industry, sanitation, biodiversity, fisheries, and a wide range of ecosystem services. However, increasing urbanisation, industrial activity, agricultural intensification, inadequate wastewater treatment, and poorly managed waste continue to place pressure on river environments. River pollution is consequently a persistent environmental-management problem affecting both high-income and developing countries. The 2024 global assessment under Sustainable Development Goal indicator 6.3.2 found that only 56% of monitored water bodies in 120 reporting countries were classified as having good ambient water quality in 2023. The assessment also identified substantial gaps in water-quality monitoring, particularly in low- and middle-income countries, limiting the ability to accurately assess progress toward pollution-reduction targets [1].

River pollution management has traditionally emphasised wastewater treatment, industrial discharge regulation, sanitation improvement, and control of identifiable pollution sources. These measures remain fundamental, but river recovery increasingly requires management approaches that integrate pollution prevention with catchment management, ecological restoration, environmental monitoring, regulatory enforcement, and institutional coordination. This transition is evident in established river-management systems. In the Rhine basin, for example, long-term cooperation among the countries sharing the basin has supported reductions in conventional pollutants, while the current Rhine 2040 programme addresses nutrients, micropollutants, sediment quality, waste and plastics, and ecological conditions. Its micropollutant strategy includes a target of reducing inputs by at least 30% by 2040 relative to 2016–2018 [2].

The Thames provides another example of long-term river management in a high-income setting. Its current river-basin management framework establishes environmental objectives and programmes of measures involving government agencies, local authorities, water companies, land managers and other stakeholders. These measures include improvements to sewage-treatment works, environmental permitting and actions addressing both point and diffuse pollution [3]. Nevertheless, current monitoring demonstrates that significant pollution pressures remain. Environment Agency data show that pollution incidents associated with Thames Water's sewerage and water-supply assets reached 523 in 2024, compared with 393 in 2023 [4].

The challenges are often more pronounced in rapidly developing river basins, where population growth, expanding urban settlements, industrialisation and limited wastewater infrastructure can increase pollutant loads. The Ganga River in India illustrates this situation at a large basin scale. The National Mission for Clean Ganga identifies municipal sewage and industrial pollution as major pressures and reports that approximately 12,000 million litres per day of sewage is generated in the Ganga basin, compared with about 4,000 million litres per day of treatment capacity. Industrial pollution, although smaller in volume, includes toxic and non-biodegradable discharges and is therefore considered particularly significant [5]. The Namami Gange Programme consequently combines sewerage-treatment infrastructure with industrial-effluent monitoring, river-surface cleaning, biodiversity conservation, afforestation, public awareness, and related river-rejuvenation measures [6].

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The Citarum River in Indonesia provides a further developing-country example in which pollution control is approached at watershed scale. The World Bank identifies untreated domestic, industrial and livestock wastewater, solid waste, watershed degradation and sedimentation among the major pressures affecting the basin. The Citarum Harum programme was established to accelerate pollution and environmental-damage control and incorporates industrial and domestic wastewater management, solid-waste management, law enforcement, land rehabilitation, public education, and water-quality monitoring [7]. Independent research nevertheless shows that water-quality conditions within the Upper Citarum remain spatially variable and that classifications differ according to the assessment method used [8].

Although the Rhine, Thames, Ganga and Citarum differ in geographical setting, pollution pressures, governance structures, infrastructure, and socioeconomic conditions, each provides documented experience of attempts to reduce river pollution and restore river environments. Existing studies and institutional assessments commonly examine individual rivers, specific pollution-control programmes, or particular restoration interventions. A comparative assessment applying consistent analytical criteria across contrasting national settings can provide additional insight into the relationship between management intervention and environmental outcome. Such an approach is important because the implementation of a programme or construction of infrastructure does not necessarily demonstrate that river condition has improved uniformly or that ecological recovery has been achieved.

Accordingly, this study compares river pollution-control and restoration approaches in the Rhine River of Germany, the Thames River of the United Kingdom, the Ganga River of India, and the Citarum River of Indonesia. The cases are examined using a common framework encompassing pollution pressures, management interventions, institutional mechanisms, documented environmental outcomes, persistent pollution, and implementation limitations. The study seeks to determine how different management approaches have addressed river pollution, what measurable improvements have been documented, and which challenges remain despite intervention. By examining contrasting environmental-management contexts within a common framework, the study contributes a common comparative framework for examining management interventions, documented environmental outcomes, persistent pollution, and implementation limitations across contrasting river-management contexts, while identifying recurring

management requirements and context-specific constraints that influence the sustainability of river restoration.

2. CONCEPTUAL FRAMEWORK FOR RIVER POLLUTION CONTROL AND RESTORATION

2.1. Major Sources and Pathways of River Pollution

River pollution arises from both point and diffuse sources operating within river basins. Municipal wastewater, industrial discharges, agricultural runoff, urban drainage, solid waste, and other human activities can introduce nutrients, pathogens, metals, pesticides, plastics, and other contaminants into receiving waters [9,10]. Point sources, such as wastewater outfalls and industrial discharge points, are comparatively identifiable and can be addressed through discharge standards, treatment requirements, and regulatory monitoring. Diffuse pollution is more spatially dispersed and may enter rivers through agricultural and urban runoff, making its identification and control more complex [9,10].

The environmental consequences depend on the type, concentration and persistence of contaminants and on the characteristics of the receiving river. Excessive nitrogen and phosphorus inputs can promote eutrophication, alter dissolved-oxygen conditions and impair aquatic ecosystems, while hazardous substances and emerging contaminants may create additional risks for organisms and water uses [11]. River pollution should therefore be considered as a combination of contaminant sources, transport pathways and environmental responses, rather than solely as the presence of individual pollutants.

2.2. Pollution-Control Approaches

River pollution control involves interventions applied at different stages of the pollution pathway. Source prevention seeks to reduce or eliminate pollutant generation through cleaner production, restrictions on hazardous substances, improved agricultural practices, sanitation and solid-waste management. Containment and treatment aim to prevent pollutants from reaching rivers or remove them before discharge, particularly through municipal and industrial wastewater-treatment systems.

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Conventional treatment can substantially reduce organic matter, nutrients and pathogens, while additional treatment and source controls may be required for selected micropollutants and other persistent contaminants [9,10].

The effectiveness of these measures also depends on regulatory and institutional arrangements. Standards, permits, monitoring, enforcement and operational maintenance determine whether pollution-control measures function as intended. Consequently, river management involves both technical intervention and institutional implementation. This study therefore considers pollution control not simply in terms of the technologies or infrastructure adopted, but also in relation to whether the interventions produce measurable improvements in river condition.

2.3. River Restoration and Ecological Recovery

Pollution control and ecological restoration are related but distinct components of river management. Reducing pollutant inputs can improve water quality, but ecological recovery may also require restoration of physical habitat and river processes. River restoration can include improvements in river connectivity, floodplain and riparian habitats, sediment processes, channel structure and biological communities [12]. Such measures seek to recover ecological functions that may remain degraded even after improvements in chemical water quality.

Restoration outcomes depend on river characteristics, restoration design, spatial scale, existing pressures and the time available for ecological responses to develop. A global review of restored streams and rivers found that nutrient-retention responses varied among restoration settings and were influenced by factors such as restoration extent and hydrological connectivity [13]. Consequently, restoration should be evaluated through both environmental condition and ecological response rather than through implementation of restoration activities alone.

2.4. Environmental Governance and Institutional Coordination

River pollution is influenced by activities distributed across entire catchments; consequently, effective management requires coordination beyond individual river reaches. Integrated Water Resources Management (IWRM) provides a framework for coordinating water, land and related resources across sectors while maintaining ecosystem sustainability [14]. Such an approach is particularly relevant where pollution sources, infrastructure and ecological impacts extend across administrative boundaries.

Environmental governance involves the interaction of regulation, monitoring, enforcement, institutional responsibilities and stakeholder participation. Under the European Water Framework Directive, river-basin management is linked to monitoring and assessment of ecological and chemical status and to measures intended to achieve environmental objectives [15]. These principles provide a conceptual basis for examining how institutional arrangements influence the implementation and continuity of river-pollution control and restoration measures.

2.5. Framework for Comparative Assessment

Based on the preceding concepts, the study applies a common framework to examine each river through five linked stages:

pollution pressures → management interventions → documented environmental outcomes → remaining pollution → implementation gaps and sustainability considerations.

Policy and institutional mechanisms are considered across these stages to determine how interventions are implemented and maintained. The framework distinguishes between the implementation of a management measure and its observed environmental outcome, allowing the study to assess whether reported interventions are accompanied by measurable changes in river condition. It also recognises that differences in socioeconomic conditions, governance arrangements, river-basin characteristics and management capacity influence the outcomes of pollution-control and restoration measures.

3. MATERIALS AND METHODS

3.1. Research Design

This study employed a comparative case-study design based on secondary evidence to examine river pollution control and restoration in four contrasting national contexts. The units of analysis were the Rhine River in Germany, the Thames River in the United Kingdom, the Ganga River in India, and the Citarum River in Indonesia. A common set of assessment criteria was applied to all four cases to facilitate systematic comparison while retaining the differences in environmental and institutional context.

3.2. Case Selection

The cases were selected purposively using four criteria: (i) substantial documented river-pollution pressures; (ii) identifiable pollution-control or restoration interventions; (iii) availability of evidence on environmental outcomes; and (iv) sufficient information to examine continuing problems and implementation limitations. Germany and the United Kingdom were classified as high-income economies, while India and Indonesia were classified as lower-middle- and upper-middle-income economies, respectively, according to the World Bank's country and lending classification [16]. The selection was intended to provide contrasting socioeconomic and institutional contexts rather than to rank countries or rivers.

3.3. Data Sources

The study used secondary documentary evidence from peer-reviewed journal articles, government agencies, river-basin organisations, international organisations, official monitoring reports, and programme documents. Priority was given to official sources for information on intervention status, infrastructure, monitoring results and regulatory measures, while peer-reviewed studies were used to provide independent assessment of environmental outcomes and continuing limitations. Recent sources were prioritised for current programme status and contemporary environmental conditions, while earlier sources were retained where they documented the historical development of major pollution-control interventions.

3.4. Data Extraction

For each river, information was organised in a common evidence matrix. The matrix recorded pollution pressures, major interventions, institutional mechanisms, documented environmental outcomes, remaining pollution, and implementation limitations. Quantitative findings were retained in their reported units and definitions. No numerical transformation was undertaken where indicators, sampling approaches or assessment methods differed between river systems.

3.5. Comparative Analysis

The extracted evidence was examined using the sequence established in the conceptual framework:

pollution pressures → management interventions → documented environmental outcomes → remaining pollution → implementation gaps and sustainability considerations.

For each case, reported management actions were distinguished from measured environmental outcomes. Particular attention was given to evidence demonstrating changes in water quality or ecological condition, rather than relying solely on programme targets, planned infrastructure or stated objectives. Findings were then compared across the four cases using common qualitative criteria and the available quantitative evidence.

3.6. Assessment of Transferability

Transferability was examined after the cross-case comparison by considering the conditions that may influence application of management approaches in other river systems. These included institutional capacity, infrastructure, financing, monitoring, regulatory enforcement, and stakeholder participation. Transferability was assessed as a question of adaptation to local conditions rather than direct replication of individual interventions.

4. FOUR RIVER CASE STUDIES

4.1. Rhine River, Germany

The Rhine provides a well-documented example of long-term river-pollution control followed by ecological restoration. A major turning point occurred after the Sandoz chemical accident of 1986, when fire-fighting water containing up to 30 tonnes of pesticides entered the Rhine and caused extensive mortality of fish and other organisms along hundreds of kilometres. Following the accident, the Rhine countries adopted the Rhine Action Programme in 1987, with coordination and assessment entrusted to the International Commission for the Protection of the Rhine (ICPR). The programme introduced stricter requirements for municipal and industrial wastewater-treatment plants, including a third treatment stage for phosphorus and nitrogen removal, while also pursuing ecological recovery of the river system [17,18].

The programme produced measurable pollution-control results. By 2000, point-source inputs of most pollutants classified as priority substances had declined by 70–100% compared with 1985, while the proportion of municipalities and industries connected to wastewater-treatment plants increased from 85% to 95%. The ICPR also reported a substantial reduction in accidents involving substances hazardous to water as industrial facilities adopted improved accident-prevention measures [19].

A specific industrial example is the BASF wastewater-treatment facility at Ludwigshafen, one of the largest wastewater-treatment plants in Europe and the largest on the Rhine. The facility treated nearly 100 million m³ of BASF production wastewater annually, in addition to approximately 20 million m³ of wastewater from surrounding municipalities. BASF also reported a 23% reduction in wastewater contamination between 2010 and 2015. The plant uses biological treatment and monitors wastewater entering and leaving the treatment system, with highly contaminated streams capable of being diverted for special treatment [20].

Pollution reduction was accompanied by ecological restoration. Under the *Rhine 2020* programme, approximately 140 km² of floodplains have been reactivated since 2000, 124 floodplain waters have been reconnected with the main river, and about 166 km of riverbanks have been ecologically upgraded. Several hundred salmon now return annually from the North Sea to accessible Rhine tributaries and reproduce naturally, while fish-passage measures have reopened migration routes at several major barrages [21].

Nevertheless, contemporary management shows that the pollution problem has not been completely resolved. The Rhine 2040 programme identifies continuing priorities including reduction of nutrient inputs, reduction of micropollutant inputs by at least 30%, further pollutant-load reduction, improvement of sediment quality, and reduction of waste and plastic inputs. It also calls for improved monitoring of substances for which standard analytical methods are not available [2]. A December 2025 ICPR interim assessment reported positive developments, particularly in wastewater treatment, but concluded that further measures across wastewater, industry and agriculture were still necessary to achieve the 2040 micropollutant-reduction target [22].

The Rhine case therefore provides a documented sequence of acute pollution event → strengthened pollution controls → measurable pollutant reduction → ecological restoration → continuing management of emerging contaminants.

4.2. Thames River, United Kingdom

The Thames provides a well-documented example of long-term recovery from severe urban and industrial pollution. Historically, untreated and inadequately treated sewage contributed to very low dissolved-oxygen concentrations in the tidal river. Improvements in sewage treatment subsequently produced measurable ecological recovery.

Andrews and Rickard documented the development of a thriving estuarine community between 1976 and 1980 in areas affected by the modernised Beckton and Crossness sewage-treatment works, linking the recovery to improved effluent quality. Their study identified increases in population size, fecundity and species diversity in previously degraded reaches [23].

A major contemporary intervention is the Thames Tideway Tunnel, a 25-km sewer constructed beneath London to intercept sewage from combined sewer overflows and transfer it for treatment rather than allowing it to enter the tidal Thames. The system provides approximately 1.6 million m³ of storage capacity and is designed to stop about 95% of the relevant sewage spills from entering the tidal Thames [24].

The intervention has produced an early measurable operational result. Tideway reported in June 2026 that, since the first connection in August 2024, the 25-km tunnel system had intercepted more than 20 million tonnes of storm sewage that would otherwise have entered the river. The system was reported to be operating at full capacity and provides more than 1.5 billion litres of storage capacity within London's sewer network [25].

An earlier and more targeted intervention is the Lee Tunnel, which has operated since 2016 and intercepts sewage previously discharged through the Abbey Mills pumping station. Evidence presented to the London Assembly reported that there had been no storm-diluted sewage discharges from Abbey Mills since the Lee Tunnel was commissioned. A December 2022 ecological study recorded 714 fish representing 12 species in the Channelsea River, while a 2023 assessment reported that water quality in the Channelsea River and River Lea could be classified as “good to excellent” under the cited Water Framework Directive classifications [26].

Despite these interventions, pollution pressures remain within the wider Thames system. Environment Agency data show that Thames Water recorded 470 category 1–3 pollution incidents from sewerage assets in 2024, compared with 271 in 2021. There were also 33 serious pollution incidents in 2024, compared with 12 in 2021. Across sewerage and water-supply assets, Thames Water recorded 523 category 1–3 incidents in 2024 [4].

The Thames case therefore provides evidence of historical ecological recovery, targeted sewage interception, measurable operational improvement, and continuing pollution pressures.

4.3. Ganga River, India

The Ganga is a highly populated river system where pollution management has required expansion of sewage-treatment infrastructure alongside industrial-effluent control and river restoration. The Namami Gange Programme combines sewerage networks and sewage-treatment plants (STPs) with industrial-effluent monitoring, river-surface cleaning, ecological conservation, and public participation [6].

A specific municipal wastewater intervention is evident in Varanasi. The NMCG project database identifies the 140 MLD Dinapur STP, operational from 1 February 2020, and the 50 MLD Ramana STP, operational from 1 November 2021. Both are listed as operational Ganga projects, with the Ramana facility implemented under the Hybrid Annuity Model (HAM). Together, the two facilities provide 190 MLD of sewage-treatment capacity [27].

Industrial pollution presents a different management challenge. The Jajmau tannery cluster in Kanpur is a major example of collective industrial wastewater management within the Ganga basin. According to the NMCG *Annual Report 2023–24*, 381 tanneries were associated with the Jajmau, Banthar and Unnao clusters: 325 were connected to the Jajmau CETP, 27 to the Banthar Leather Technology Park, and 15 to the Unnao facility. The same assessment reported that the Jajmau and Banthar CETPs were non-compliant with notified effluent-discharge standards, while the Unnao CETP was non-operational. Among eight CETPs assessed on the main stem of the Ganga during April–December 2023, one was compliant, one was non-operational, and six were non-compliant [28].

Evidence of water-quality change is available from NMCG's comparison of monitoring data between 2014 and 2023. Median biochemical oxygen demand improved at 44 of 53 comparable locations. Median pH and dissolved oxygen met the applicable bathing-water criteria at all 53 comparison locations in both years, while biochemical oxygen demand and faecal-coliform conditions varied among individual river stretches [28].

The Ganga case therefore contains both measurable progress and continuing implementation problems. Expansion of sewage-treatment infrastructure and improvement in selected water-quality indicators are documented, while the CETP findings demonstrate that treatment infrastructure does not necessarily ensure continuous compliance with discharge standards [28].

4.4. Citarum River, Indonesia

The Citarum River is managed through a combination of watershed intervention, wastewater control, solid-waste management, regulatory enforcement, and institutional coordination. The basin is important for water supply, irrigation, industry, and hydropower, but has been affected by untreated domestic and industrial wastewater, livestock waste, solid waste, watershed degradation, and sedimentation. The Citarum Harum programme was established under Presidential Regulation No. 15/2018 to accelerate pollution and environmental-damage control throughout the watershed. Its action plan covers industrial and domestic wastewater, livestock waste, solid waste, critical land, spatial management, law enforcement, environmental education, and water-quality monitoring [7].

An earlier intervention was the West Tarum Canal Water Quality Management Project, supported by the Asian Development Bank. Completed in 2008, the project established a comprehensive water-quality monitoring network, developed a water-quality database, identified suspected pollution sources, strengthened the capacity of the basin-management organisation, and promoted stakeholder participation and improved water-resource management [29].

Specific enforcement activities under Citarum Harum provide evidence of direct control of industrial pollution. In Sector 7, 32 industrial facilities were subject to routine monitoring, normally three times annually, with additional inspections where wastewater-management problems were identified. In 2024, one facility without an adequate wastewater-treatment installation was sanctioned and referred to the police [30]. In a separate case in Kampung Cilisung, Sukamenak, authorities found inadequate wastewater treatment and discharge openings leading toward a Citarum tributary. After repeated warnings in April and July 2024, the discharge openings were blocked in October to prevent direct release, and the company was instructed to install a wastewater-treatment system meeting the required standards [31].

Solid-waste removal provides another measurable intervention. Official reporting from Citarum Harum Sector 6 recorded approximately 49,971 m³ of river waste removed during 2024, equivalent to about 91% of the stated annual target of 54,700 m³. Domestic waste collection reached 17,826 m³, or 167% of the stated target of 10,700 m³ [32].

At the river level, West Java authorities reported that the Citarum Water Quality Index increased from 50.78 in 2023 to 51.05 in 2024, with the river classified locally as lightly polluted [33]. Independent assessment indicates that water-quality conditions remain spatially variable. A 2025 study of the Upper Citarum found classifications ranging from good to poor depending on the index applied, while the Pollution Index classified locations from moderately to severely polluted; downstream conditions were generally poorer than upstream conditions [8].

The Citarum case therefore provides evidence of watershed-scale management combined with targeted industrial enforcement, monitoring, and waste-removal activities, while independent assessment continues to show variation in water-quality conditions across the river system.

5. COMPARATIVE ASSESSMENT

The four case studies reveal distinct combinations of pollution-control, wastewater-treatment, regulatory, and restoration measures. The comparison considers the nature of the interventions, their implementation mechanisms, documented environmental outcomes, and continuing pollution pressures. Because the four river basins differ in monitoring systems, indicators, geographical scale, and reporting periods, the findings are compared using common qualitative criteria together with the quantitative evidence available for each case rather than through a single composite performance measure.

5.1. Comparative Pollution-Control Interventions

The four cases represent different configurations of river-pollution management. The Rhine is characterised by long-term transboundary basin governance, supported by municipal and industrial wastewater controls and ecological restoration. Its management has progressively expanded from conventional pollution control to include micropollutants, nutrients, plastics, and other emerging pressures [2,19].

The Thames combines regulatory river-basin management with large-scale urban wastewater infrastructure. Its interventions include environmental permitting and major systems for intercepting sewage from combined sewer overflows [3,25,26].

The Ganga reflects a large-scale national pollution-abatement approach centred on sewage-treatment infrastructure and industrial-effluent control, supported by monitoring and enforcement under the Namami Gange Programme [6].

The Citarum follows a broader watershed-management approach integrating wastewater control, solid-waste management, monitoring, industrial inspection, enforcement, and watershed activities [7,30–33].

Overall, the cases differ in the scale and configuration of intervention. The Rhine emphasises transboundary coordination, the Thames combines regulation with urban sewerage infrastructure, the Ganga emphasises expansion of pollution-control capacity and industrial regulation, and the Citarum combines watershed management with direct enforcement and waste control.

5.2. Comparative Environmental Outcomes

The documented environmental outcomes differ in both type and scale of evidence. The Rhine shows long-term reductions in conventional pollutant inputs accompanied by ecological restoration, including improvements in floodplain connectivity and riverbank condition [19,21].

The Thames provides evidence of historical ecological recovery and reductions in specific sewage-discharge pathways following wastewater-treatment improvements and sewerage interventions [24–27].

For the Ganga, NMCG monitoring data for 2014–2023 reported improvement in median biochemical oxygen demand at 44 of 53 comparable locations, while median pH and dissolved oxygen met applicable bathing-water criteria at all 53 locations [28].

The Citarum provides both programme-level and independent evidence. West Java reporting recorded an increase in the Water Quality Index from 50.78 in 2023 to 51.05 in 2024, while independent assessment of the Upper Citarum found spatial variation in water-quality classification and poorer conditions at downstream locations [8].

Overall, the four cases provide evidence of environmental improvement, but the indicators, spatial coverage, monitoring methods, and assessment periods differ. The results therefore cannot be treated as directly equivalent measurements of river recovery.

5.3. Comparative Remaining Pollution

The four rivers continue to experience different forms of residual pollution. The Rhine faces micropollutants, nutrients and plastics [2,22], while the Thames continues to experience wastewater-related pollution incidents [4].

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The Ganga remains affected by sewage and industrial effluents [5], while the Citarum continues to experience wastewater, solid waste and sedimentation pressures [7,8,33]. Thus, pollution-control efforts have reduced some pressures but have not eliminated all sources of river contamination.

5.4. Comparative Implementation Gaps

Implementation gaps across the four cases relate mainly to the continuity, enforcement, and operational effectiveness of pollution-control measures. The evidence indicates that regulations, treatment facilities, monitoring systems, and restoration programmes require continued implementation and oversight to maintain environmental performance. The nature of the gaps varies among the four river systems according to their pollution pressures and institutional conditions.

5.5. Cross-Case Comparison

Across the four cases, pollution-control interventions are associated with documented improvements, but residual pollution and implementation gaps remain. The comparison also distinguishes management outputs such as treatment capacity, sewage interception, waste removal, and restoration activities from measured environmental responses.

Because monitoring approaches and environmental indicators differ among the four rivers, the results do not support a single numerical measure of comparative river recovery. The evidence instead demonstrates the need to assess intervention, environmental response, and remaining pollution together.

6. DISCUSSION

The comparative assessment indicates that river pollution control is a continuing environmental-management process rather than a one-time intervention. Across the four cases, substantial management efforts have produced measurable improvements, but persistent pollution and uneven environmental responses remain. The findings suggest that the effectiveness of river restoration depends not only on the type of intervention adopted but also on its continuity, implementation, monitoring, and institutional support.

6.1. Integrated River-Basin Management

The cases demonstrate the importance of managing pollution at the river-basin scale. Pollution sources often originate beyond the immediate river channel and may include municipalities, industries, agricultural land and waste-management systems. Consequently, effective restoration requires coordination across administrative boundaries and between sectors. The comparison indicates that basin-level planning provides a broader basis for connecting pollution control with ecological restoration and long-term water-quality management.

6.2. Linking Management Actions with Environmental Outcomes

An important finding is the distinction between management outputs and environmental outcomes. Treatment plants constructed, sewage intercepted, waste removed, or facilities monitored are indicators of implementation, but they do not by themselves establish ecological recovery. Environmental effectiveness must be assessed through changes in water quality and, where available, ecological indicators. The differences in the evidence reported for the four rivers demonstrate the importance of evaluating interventions through measurable environmental responses rather than programme implementation alone.

6.3. Changing Nature of River-Pollution Challenges

The comparison also indicates that river-pollution priorities can change as conventional pollution is brought under greater control. Persistent and emerging contaminants, plastics, diffuse pollution, infrastructure-related discharges and other pressures may become increasingly important as management progresses. This creates a need for adaptive environmental management, in which monitoring results inform revisions to pollution-control priorities rather than maintaining a fixed intervention strategy over long periods.

6.4. Governance, Compliance and Operational Performance

Environmental regulations and infrastructure are effective only when they are supported by appropriate implementation and oversight. The evidence from the four cases shows the importance of regulatory compliance, routine monitoring, enforcement, operation and maintenance, and institutional continuity.

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These factors influence whether pollution-control measures function as intended over time. The comparison therefore suggests that environmental governance should be assessed alongside technological investment when evaluating river-restoration programmes.

6.5. Sustainability and Transferability

The findings do not support direct replication of one country's river-management model in another setting. Differences in institutional capacity, infrastructure, financing, environmental conditions and administrative arrangements affect implementation. Nevertheless, several management principles recur across the cases: pollution prevention at source, effective wastewater treatment, continuous monitoring, regulatory enforcement, institutional coordination and ecological restoration. These principles may provide a basis for adaptation, while the specific design of interventions should reflect local conditions.

6.6. Limitations and Research Gaps

This study relies on secondary evidence and therefore does not independently verify reported water-quality measurements or programme outcomes. Differences in monitoring methods, indicators, assessment standards and periods of observation also limit direct numerical comparison. In addition, improvements in river condition cannot always be attributed to individual interventions because multiple measures may operate simultaneously. The selection of four purposively chosen rivers further limits generalisation.

Several areas warrant further investigation. Greater attention is needed to the relationship between pollutant reduction and ecological recovery, particularly over longer periods. Monitoring of emerging and diffuse contaminants also requires strengthening, as does evaluation of wastewater-treatment performance after infrastructure becomes operational. Future research should further integrate climate-related pressures, long-term financing, institutional performance and social participation into assessments of river-restoration sustainability.

7. CONCLUSION

This study compared river pollution-control and restoration approaches in the Rhine, Thames, Ganga, and Citarum using a common environmental-management framework. The four cases demonstrate that sustained pollution-control and restoration efforts can produce measurable improvements in river systems, but that progress does not necessarily result in complete or uniform ecological recovery.

The comparison shows that the nature of the remaining environmental challenge differs among the rivers. While established management systems have reduced several conventional pollution pressures in the Rhine and Thames, emerging contaminants, wastewater-related incidents, and other pressures remain. In the Ganga and Citarum, expansion of pollution-control infrastructure and broader watershed interventions have contributed to documented improvements, while wastewater management, industrial compliance, solid waste, and spatial differences in river condition remain important concerns.

A common finding across the cases is the importance of continuous environmental management. Pollution prevention, wastewater treatment, monitoring, regulatory enforcement, institutional coordination, and ecological restoration need to operate together rather than as isolated measures. The comparison also demonstrates that implementation of infrastructure or programmes should be accompanied by assessment of actual changes in water quality and ecological condition.

No single river-restoration model can be directly applied across different national contexts. Differences in pollution sources, governance, infrastructure, financial capacity, monitoring systems, and environmental conditions require locally appropriate approaches. Nevertheless, the four cases indicate that long-term progress depends on sustained institutional commitment, effective implementation, and adaptation to changing pollution pressures.

The comparative framework provides a practical basis for assessing river-pollution control by considering management interventions together with environmental outcomes, persistent pollution and implementation constraints, and may support the planning and evaluation of river-restoration programmes in different river-basin contexts.

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The study is limited by its reliance on secondary evidence, differences in monitoring approaches and indicators, and the purposive selection of four river systems. Future research should strengthen long-term assessment linking pollution reduction with ecological recovery and should give greater attention to emerging contaminants, diffuse pollution, climate pressures, and the operational performance of pollution-control infrastructure.

Overall, sustainable river restoration requires a long-term, integrated, and evidence-based management approach in which environmental interventions are continuously evaluated against measurable improvements in river health.

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